Production Order Create/Edit — Detailed Design Document (詳細設計書)

> **Revision of 2026-10-24 (WI-017).** This file supersedes `001_DD_製造指示登録・編集.md`, which stays unchanged as the earlier approved revision. Change: the production-order notes limit is 1000 characters (REQ-090, supersedes the 500 of REQ-016); see `work-items/WI-017/`.

001_DD — implements 001_BD (SCR-001), requirements REQ-010–REQ-019.

Document control (改版履歴)

Field	Value
Document ID	001_DD
Category	UI + API
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-002
Implements	001_BD revision 10 (`001_BD_製造指示登録・編集_20261024.md`)
Created by	Claude (for ThanhTN)
Created date	2026-09-18
Last updated by	Claude (for ThongTM)
Last updated date	2026-10-24

Version	Date	Author	Revision content
1	2026-09-18	Claude (for ThanhTN)	Initial creation
2	2026-09-18	Claude (for ThanhTN)	Added the function-design (001_DD-FN) and screen-processing-design (001_DD-SPD) companions; modules 3–5 and flows P-01–P-05 moved there, with pointers left here
3	2026-09-18	Claude (for ThanhTN)	Aligned with implementation (plan revision 2): `AllowedNext()` extension method; OpenTelemetry instrumentation set; test-plan IDs filled in

Version	Date	Author	Revision content
4	2026-09-22	Claude (for ThanhTN)	Completion tracking (WI-004 REQ-033, 001_BD revision 6, 003_DB): module 1 sets `CompletedAtUtc` on `InProgress → Completed`; state-transition side effect, database mapping and test viewpoints updated
5	2026-09-23	Claude (for ThanhTN)	Diagrams redrawn for the revised DD template (RFC 0009): the layout sketch and the discard dialog as SVG wireframes under `wireframes/`; the order status transitions also drawn as a Mermaid state diagram above the table. No field, rule, API or processing changes
6	2026-09-23	Claude (for ThanhTN)	WI-005: the UI is Japanese. The message catalog lists the Japanese text with an English gloss (WI-005 DEC-003); status options, lock hint, required legend, timestamp format (`YYYY/MM/DD HH:mm`, WI-005 DEC-006), `lang="ja"`, wireframes and mockup updated. A Japanese edition of the mockup was added (`mockups/001_DD_screen-a-mockup.ja.html`). No field, rule, API or processing changes
7	2026-10-24	Claude (for ThongTM)	WI-017 (REQ-090–REQ-092): notes limit 500 → 1000 characters. Implements 001_BD revision 10. Request DTO and client checks (≤ 1000), items (11)/(12) (counter `{n}/1000`, polite announcement from 900, DEC-003), the MSG-E006 text and the notes test viewpoint updated; layout wireframe and EN/JA mockups replaced by dated copies. No other field, rule, API or processing change. Dated file name per WI-017 DEC-002

Overview and reference documents (概要・目次)

Field	Value
File / component name	Frontend `ProductionOrderPage.tsx` (+ feature components); backend `ProductionOrdersController`, `ProductsController`, `ProductionOrderService`, `ProductionOrder` entity
Overview	Create/edit form for one production order, with status transitions, field locking, stale-save protection and discard confirmation, backed by a JSON API over the 001_DB schema

Module / method / processing index

No	Name	Overview	Notes
1	`ProductionOrder` (Domain)	Entity + business invariants: transitions, field lock, quantity/notes rules	Unit-tested without a host (0001_ADR)
2	`ProductionOrderStatus` (Domain)	Enum + allowed-next-states table	Single source of M-02 / V-06
3	`ProductionOrderService` (Application)	Create / get / update use cases, validation orchestration, error mapping	Designed in 001_DD-FN
4	`IPlantClock` / `PlantClock`	"Today" and current year in the plant timezone	Designed in 001_DD-FN
5	`IOrderNumberIssuer` / `OrderNumberIssuer` (Infrastructure)	Per-year sequence upsert	Designed in 001_DD-FN
6	`ProductionOrdersController`, `ProductsController` (Api)	HTTP endpoints, auth policy, Problem Details	Contract in 001_DD-API
7	`ProductionOrderPage` (frontend)	Route component: load, mode, states, title	

No	Name	Overview	Notes
8	`ProductionOrderForm` (frontend)	Fields, client validation, dirty tracking, save	
9	`DiscardChangesDialog` (frontend)	Confirmation on Cancel with unsaved changes	REQ-019
10	`apiClient` (frontend)	JSON fetch wrapper, Problem Details parsing, 401 handling	Shared
11	Processing flows P-01–P-05	Load, create, update, cancel, error handling	Designed in 001_DD-SPD

Reference documents

No	Document	Purpose / use	Notes
1	`docs/en/010_basic-design/001/001_BD_製造指示登録・編集_20261024.md`	Screen items, validation V-01–V-08, events E-01–E-09, flows	Item numbers (n) below are 001_BD §3 numbers
2	`docs/en/database/001/001_DB_製造指示登録・編集_20261024.md` (001_DB)	Tables, constraints, counter upsert, `xmin`, privileges	
3	`docs/en/020_detailed-design/001/001_DD-API_製造指示登録・編集_20261024.md` (001_DD-API)	Endpoint request/response/error catalog	Companion
3a	`docs/en/020_detailed-design/001/001_DD-FN_製造指示登録・編集_20261024.md` (001_DD-FN)	Shared service/clock/issuer method design	Companion
3b	`docs/en/020_detailed-design/001/001_DD-SPD_製造指示登録・編集_20261024.md` (001_DD-SPD)	Per-component processing flows	Companion
4	`docs/en/architecture/0001/0001_ADR_backend-layered-structure.md`	Layer placement	
5	`docs/en/architecture/0002/0002_ADR_auth-rbac-foundation.md`	Cookie auth, fallback policy, CSRF (DEC-020)	
6	`work-items/WI-002/brief.md`, `decisions.md`	REQs, DEC-001–DEC-024	

Referenced by

No	Document	Purpose / use	Notes
1	001_DD-API, 001_DD-FN, 001_DD-SPD	Companions; each lists 001_DD as its parent	
2	`work-items/WI-002/evidence.md`	Traceability	

Component / file organization

X-1. Path structure

No	Path / namespace	Purpose	Notes
1	`src/backend/ProductionManagementAI.Domain/ProductionOrders/`	`ProductionOrder`, `ProductionOrderStatus`, `Product`, `DomainRuleViolation`	No external dependencies
2	`src/backend/ProductionManagementAI.Application/ProductionOrders/`	`ProductionOrderService`, request/response DTOs, `IPlantClock`, `IOrderNumberIssuer`, `IProductionOrderRepository`, `ProductionOrderTelemetry`	
3	`src/backend/ProductionManagementAI.Infrastructure/ProductionOrders/`	EF configuration, repository, `OrderNumberIssuer`, `PlantClock`; new migration `AddProductionOrders`	
4	`src/backend/ProductionManagementAI.Api/Controllers/`	`ProductionOrdersController`, `ProductsController`	
5	`src/frontend/src/features/production-orders/`	`ProductionOrderPage.tsx`, `ProductionOrderForm.tsx`, `DiscardChangesDialog.tsx`, `validation.ts`, `messages.ts`, `api.ts`, `types.ts`	
6	`src/frontend/src/lib/apiClient.ts`	Shared fetch wrapper	New
7	`tests/backend/ProductionManagementAI.Application.Tests/ProductionOrders/`	Domain + service unit tests	
8	`tests/integration/ProductionManagementAI.Integration.Tests/ProductionOrders/`	Endpoint tests (Testcontainers Postgres)	
9	`src/frontend/tests/unit/production-orders/`	Component tests (Vitest + RTL)	

X-2. Shared/common components used

No	Name	Purpose	Notes
1	`useAuth` / `ProtectedRoute` (WI-001)	Signed-in user and roles; redirect to `/login`	Existing
2	`apiClient`	`getJson` / `sendJson`: same-origin credentials, `Content-Type: application/json`, parses RFC 9457 bodies into `ApiProblem`, and on 401 clears the auth state and navigates to `/login`	New; the WI-001 auth calls keep their current code
3	ASP.NET Core `AddProblemDetails()` + `[ApiController]`	Problem Details for all API errors	New registration in `Program.cs`

X-3. Feature-level components used

No	Name	Purpose	Notes
1	`ProductionOrderForm`	Items 6–16	Stateless about routing; receives initial values + callbacks
2	`DiscardChangesDialog`	Items 17–19	Native `<dialog>` with `showModal()` (built-in focus trap, Escape)
3	`FieldError`	Inline error text linked by `aria-describedby`	
4	`MessageBanner`	Item 5, `role="status"` (success) / `role="alert"` (error)	

X-4. External APIs used — see "APIs used" below.

X-5. Responsive composition — one component tree with Tailwind responsive classes: the `sm:` breakpoint (640px) switches quantity and due date to two columns, and buttons to inline right-aligned. Below 640px everything is one column, and the buttons are full width with Save above Cancel (DOM order Cancel, Save; `flex-col-reverse` on mobile keeps the tab order logical).

Companion design documents

No	Document	Type	Covers
1	`001_DD-API_製造指示登録・編集_20261024.md`	api-design	`GET /api/products`, `POST /api/production-orders`, `GET` / `PUT /api/production-orders/{id}`
2	`001_DD-FN_製造指示登録・編集_20261024.md`	function-design	`ProductionOrderService` (list/get/create/update), `ProductionOrderMapper`, `IPlantClock`, `IOrderNumberIssuer`
3	`001_DD-SPD_製造指示登録・編集_20261024.md`	screen-processing-design	P-01–P-05 per component: page, form, save-response handling, discard dialog, controller boundary

Task / design index

No	Category	File-level task	Function/process-level task	Item-level task	Confirmed	Issue	Reviewer	Reworked	Date	Notes
1	basic-design	001_BD	SCR-001 all sections	—	yes	—	ThanhTN	yes	2026-09-18	revision 4 (V-02 upper bound)
2	detailed-design	001_DD	Modules 1–10, P-01–P-05	Items 4–19	yes	—	ThanhTN	yes	2026-09-18	this document; approved 2026-09-18
3	api-design	001_DD-API	4 endpoints	request/response fields	yes	—	ThanhTN	no	2026-09-18	approved 2026-09-18

No	Category	File-level task	Function/process-level task	Item-level task	Confirmed	Issue	Reviewer	Reworked	Date	Notes
5	function-design	001_DD-FN	7 methods	—	yes	—	ThanhTN	no	2026-09-18	approved 2026-09-18
6	screen-processing-design	001_DD-SPD	5 processing blocks	—	yes	—	ThanhTN	no	2026-09-18	approved 2026-09-18
4	detailed-design	001_DB	Tables, counter, `xmin`	—	yes	—	ThanhTN	no	2026-09-18	reviewed by user

Module design

1. `ProductionOrder` (Domain entity)

Field	Value
Description	Aggregate for one production order. Owns every invariant that doesn't need I/O: status transitions, the product/quantity lock, and quantity/notes value rules
Return type	— (entity)
Created by / date	Claude / 2026-09-18
Last modified by / date	—

Preconditions: none.

Rule type: business invariant.

Rule references

No	Rule / validator	Location	Notes
1	`ProductionOrderStatusExtensions.AllowedNext`	Domain	V-06

Condition / data references

No	Key	Source	Notes
1	Saved `Status`, `ProductId`, `Quantity`, `DueDate`	Entity state loaded by the service	V-04 "changed", V-07

Check parameters

No	Name	Type	Target field
1	`quantity`	`int`	(9)
2	`notes`	`string?`	(11)
3	`newStatus`	`ProductionOrderStatus`	(7)
4	`productId` , `quantity` vs. saved	`Guid` , `int`	(8)(9)

Arguments (factory and mutator)

No	Type	Name	Description
1	`Guid`	`productId`	`Create` / `Update`
2	`int`	`quantity`	1–999,999,999 (V-02, DEC-024)
3	`DateOnly`	`dueDate`	Date checks against "today" are done in the service (they need `IPlantClock`)
4	`string?`	`notes`	Trimmed; empty → `null` ; ≤ 1000 Unicode code points (V-05)
5	`ProductionOrderStatus`	`status`	`Update` only
6	`short` , `int`	`orderYear` , `orderSeq`	`Create` only, from `IOrderNumberIssuer`

Dependencies: none.

Processing overview: `ProductionOrder.Create(...)` returns a new `Draft` order. `Update(...)` first checks the lock and the transition, and throws `DomainRuleViolation` with a code (`MSG-E007` , `MSG-E008`) before changing any state. Value-rule failures (quantity, notes) are reported by the static `Validate...` helpers, which return error lists rather than throwing. The service calls them first so all field errors can be returned together.

Processing flow (`Update`)

Step	Description	Calls
1	If `Status != Draft` and (`productId != ProductId` or `quantity != Quantity`) → throw `DomainRuleViolation("MSG-E008")` ; nothing changed (DEC-007)	—
2	If `status != Status` and `status ∉ AllowedNext(Status)` → throw `DomainRuleViolation("MSG-E007")`	`ProductionOrderStatusExtensions.AllowedNext`
3	Assign `ProductId` , `Quantity` (only possible while `Draft`), `DueDate` , `Notes` , `Status`	—
4	Set `UpdatedAtUtc` = the given UTC time	—

Step	Description	Calls
5	If the status moved from `InProgress` to `Completed` in this call, set `CompletedAtUtc` = the same UTC time. Never set otherwise, never cleared; there is no public setter or request field for it (WI-004 REQ-033, 003_DB `ck_production_orders_completed_at_matches_status`)	—

Return value: none — state change or exception.

2. `ProductionOrderStatus` (Domain enum)

Field	Value
Description	`Draft`, `InProgress`, `Completed`, `Cancelled`, plus the `AllowedNext()` extension method in `ProductionOrderStatusExtensions` (C# enums can't hold methods), whose table is: Draft → {InProgress, Cancelled}; InProgress → {Completed, Cancelled}; Completed, Cancelled → {}
Return type	`IReadOnlySet<ProductionOrderStatus>`

Not applicable — a plain lookup table, not a rule/validator with its own fields. Persisted as a string via EF value conversion (DEC-015). The API returns `allowedNextStatuses` so the frontend never duplicates the table (M-02).

3–5. `ProductionOrderService`, `IPlantClock`, `IOrderNumberIssuer` (shared)

These modules will be reused by Screen B, so per `ai/templates/detailed-design.md` they are designed in the function-design companion, **001_DD-FN** (`001_DD-FN_製造指示登録・編集_20261024.md`). That document covers the methods, arguments, the fixed check order, telemetry points and value mapping. It is not repeated here.

6. Controllers (Api)

Field	Value
Description	`ProductionOrdersController` (`api/production-orders`) and `ProductsController` (`api/products`), both `[Authorize(Policy = "ProductionOrderEditor")]`, where the policy is `RequireRole("Admin", "Operator")` (DEC-001). Map `Result` to 200/201/400/404/409/422 Problem Details. `[Consumes("application/json")]` on POST/PUT (DEC-020)
Return type	`ActionResult<T>`

Not applicable — plain handler. Route constraint `{id:guid}`, so a non-GUID id returns 404 without reaching the service.

7. `ProductionOrderPage` (frontend)

Field	Value
Description	Route component for `/production-orders/new` and `/production-orders/:id`. Decides the mode, runs P-01, holds page state (`loading` | `ready` | `notFound` | `forbidden` | `loadError`), sets `document.title` (0-2-1), shows the banner, and renders `ProductionOrderForm`
Return type	JSX

Not applicable — plain component. Role check (UX only; the server is authoritative): if `user.roles` includes neither `Admin` nor `Operator`, render the forbidden panel (MSG-E012) without calling the API.

8. `ProductionOrderForm` (frontend)

Field	Value
Description	Controlled form for items 6–16. Keeps `initialValues` (a snapshot from load or save) and `values`; `isDirty = !equal(values, initialValues)` (with notes compared after trimming). Runs field validators on blur (E-02) and all of them on Save (E-04)
Return type	JSX

Rule type: format check (client-side, advisory; mirrors V-01–V-05 in `validation.ts`, same message IDs).

Check parameters

No	Name	Type	Target field
1	`productId`	string	(8) non-empty
2	`quantity`	string	(9) `^[0-9]{1,9}$` and ≥ 1
3	`dueDate`	`YYYY-MM-DD`	(10) non-empty; ≥ browser-local today when creating or changed (DEC-021)
4	`notes`	string	(11) `[...notes].length ≤ 1000`

Not applicable for V-06/V-07/V-08: the UI prevents V-06 and V-07 (restricted options, read-only fields), and V-08 is server-only.

9. `DiscardChangesDialog` (frontend)

Field	Value
Description	Items 17–19. Native `<dialog>` opened with `showModal()`; initial focus on **編集を続ける** (Keep editing) (`autofocus`); Escape/`cancel` event = Keep editing; on close, focus returns to the 「キャンセル」 (Cancel) button
Return type	JSX

Not applicable — plain component.

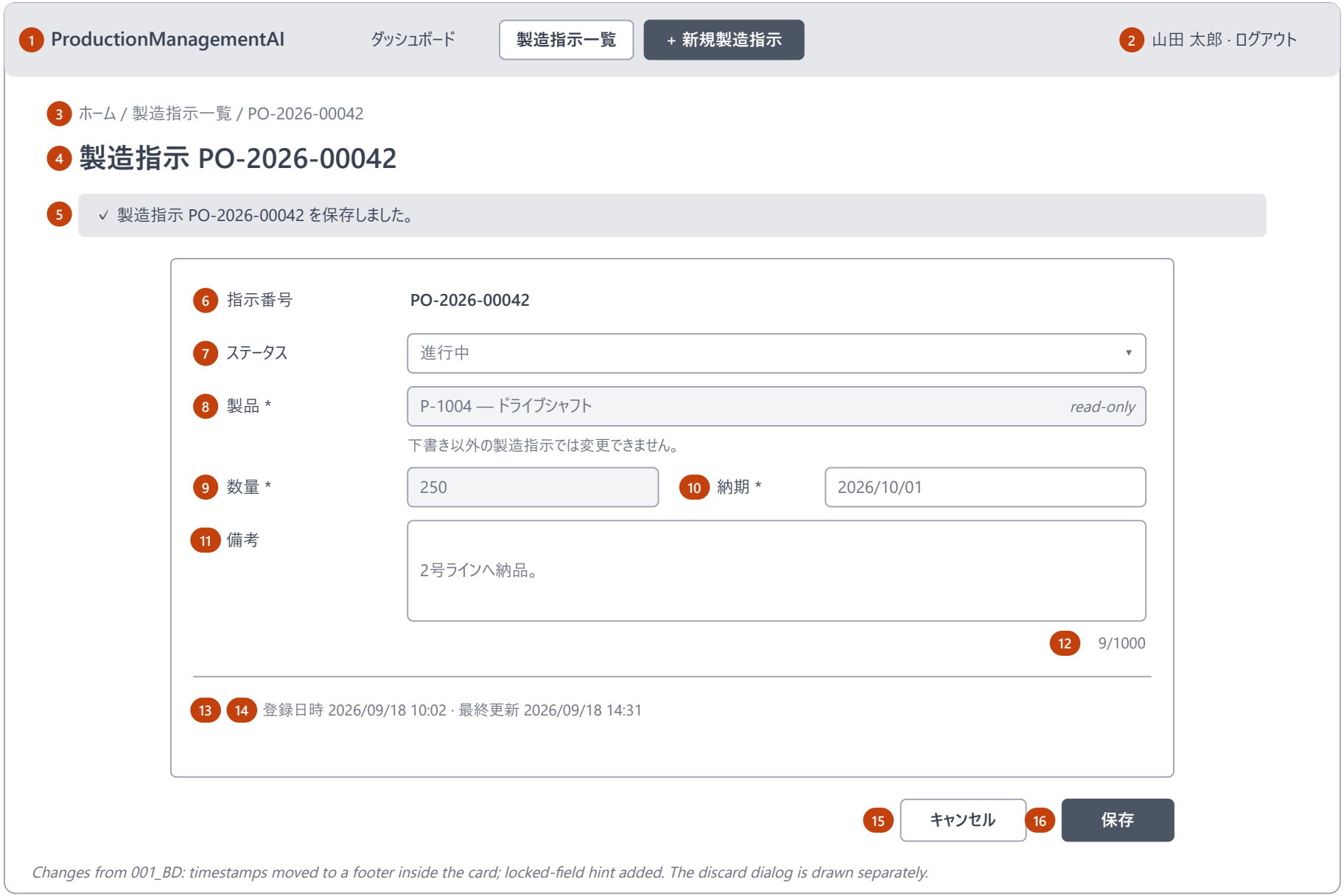
10. `apiClient` (frontend)

Field	Value
Description	`getJson<T>(url)`, `sendJson<T>(method, url, body)`. Always sends `credentials: 'same-origin'`, plus `Content-Type: application/json` on bodies. Non-2xx → throws `ApiError { status, problem?: ProblemDetails }`. 401 → calls the auth context's `onUnauthorized` (clears the user; `ProtectedRoute` then redirects to `/login`)
Return type	`Promise<T>`

Not applicable — plain module.

Screen layout and mockup

Refines the 001_BD §1 wireframe. Changes from BD: the timestamps (13)(14) moved to a footer line inside the card; the locked-field hint text was added; the discard dialog (17–19) is drawn.



Discard dialog (17–19), centered over a dimmed page:



Mockup source (revision of 2026-10-24, counter `/1000`; not published as an Artifact because WI-017 is local only, DEC-001 — the earlier Artifact https://claude.ai/artifact/FEo1RG27UjZ6vxFCUjxoHq still shows `/500`): `mockups/001_DD_screen-a-mockup_20261024.html`. It has 11 static states: create empty, product list open (30 items), validation errors, created → edit (Draft), in progress (locked), completed (terminal, overdue), stale save (409), discard dialog, not found, SP create, SP locked.

Japanese edition, with the page captions in Japanese too (WI-005), revision of 2026-10-24, local source only (the earlier Artifact https://claude.ai/artifact/QY4W3yuMnnBXxBkN2XioiD still shows `/500`): `mockups/001_DD_screen-a-mockup_20261024.ja.html`.

Region	Contains (field/control)	Notes
Header	(1)(2)	Shared app header (new small `AppHeader` component, also used on the home page)
Page head	(3)(4)(5)	Breadcrumb collapses to 「‹ 製造指示一覧」 (‹ Production orders) below 640px
Form card	(6)–(14)	`max-w-3xl` centered, `rounded-lg border border-gray-200 p-6` (same vocabulary as `LoginPage`)
Actions	(15)(16)	Right-aligned ≥ 640px; full width, stacked, 「保存」 (Save) first < 640px
Dialog	(17)–(19)	`<dialog>` with `::backdrop` dimming

Visual vocabulary follows the existing `LoginPage.tsx`: gray-900 primary button, `border-gray-300` inputs, `text-red-600` errors. Additions: `bg-gray-50` read-only fields with a lock hint, a green success banner (`bg-green-50 text-green-800`) and a red error banner (`bg-red-50 text-red-800`). Focus rings are `focus-`

`visible:ring-2 ring-gray-900 ring-offset-2`, and all text/background pairs are ≥ 4.5:1 contrast.

Screen item definition

Field	Type	Required	Validation rule	Source (BD ref)
(6) `orderNumber`	string, response only	—	—	§3 item 6
(7) `status`	enum select; label in create mode	edit: yes	Options = current + `allowedNextStatuses` (from API); select disabled when that list is empty	§3 item 7, M-01, M-02, V-06
(8) `productId`	GUID string, `<select>` of 30 products ordered by SKU, label `{sku} – {name}` (e.g. `P-1004 – ドライブシャフト`), first option 「製品を選択」 (Select a product)	yes	Non-empty (MSG-E001); server: exists (MSG-E002). `disabled`-looking but focusable read-only when not Draft (`aria-readonly`, `aria-describedby` → lock hint)	§3 item 8, V-01, V-07, DEC-022
(9) `quantity`	`<input type="text" inputmode="numeric" maxlength="9">`	yes	`^[0-9]+$`, 1–999,999,999 (MSG-E003 / MSG-E010); read-only when not Draft	§3 item 9, V-02, DEC-024
(10) `dueDate`	`<input type="date">`	yes	Non-empty (MSG-E004); ≥ today on create or when changed (MSG-E005). Client uses the browser-local date; the server uses the plant date (DEC-021)	§3 item 10, V-03, V-04
(11) `notes`	`<textarea rows="4">`	no	≤ 1000 code points (MSG-E006); no `maxlength` attribute (it counts UTF-16 units), so the counter turns red and blocks Save instead	§3 item 11, V-05
(12) counter	text	—	`{n}/1000`, red at > 1000, `aria-live="polite"` announced only at ≥ 900	§3 item 12
(13)(14) `createdAt`, `updatedAt`	ISO UTC → `formatTimestamp` (`src/lib/format.ts`: `Intl.DateTimeFormat('ja-JP', …)`, 24-hour) → 「登録日時 2026/09/18 10:02」 / 「最終更新 …」, browser timezone	—	—	§3 items 13–14, M-04
(hidden) `version`	uint	update: yes	Echoed back on PUT	V-08

Message catalog (`messages.ts`; the server returns the same IDs as `code`). The other UI strings live in the same module's `labels` object (WI-005 DEC-007):

ID	Text (Japanese, as implemented)	English gloss
MSG-E001	製品を選択してください。	Select a product.
MSG-E002	選択した製品は存在しません。	The selected product no longer exists.
MSG-E003	1以上の整数を入力してください。	Enter a whole number of 1 or more.
MSG-E004	納期を入力してください。	Enter a due date.
MSG-E005	納期に過去の日付は指定できません。	Due date can't be in the past.
MSG-E006	備考は1000文字以内で入力してください。	Notes can't exceed 1000 characters.
MSG-E007	このステータス変更は許可されていません。	This status change isn't allowed.
MSG-E008	下書き以外の製造指示では、製品と数量を変更できません。	Product and quantity can't be changed after the order leaves Draft.
MSG-E009	この製造指示は他のユーザーによって変更されました。再読み込みして最新の内容を確認してください。	This order was changed by someone else. Reload to see the latest version.
MSG-E010	数量は999,999,999以下で入力してください。	Quantity can't exceed 999,999,999.
MSG-E011	この製造指示は存在しません。	This production order doesn't exist.
MSG-E012	製造指示を管理する権限がありません。	You don't have permission to manage production orders.
MSG-E013	問題が発生しました。もう一度お試しください。	Something went wrong. Try again.
MSG-E014	利用可能な製品がありません。	No products available.
MSG-I001	製造指示 {orderNumber} を登録しました。	Production order {orderNumber} created.
MSG-I002	製造指示 {orderNumber} を保存しました。	Production order {orderNumber} saved.

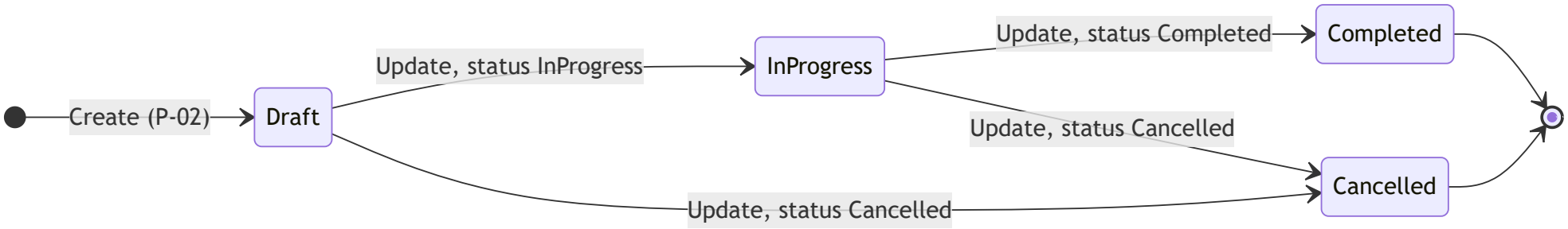
Loading / empty / error / success states

State	Trigger	UI behavior	Data shown
Loading	P-01 in flight	Header, heading placeholder and a skeleton card (`aria-busy="true"`); no form controls	—
Ready — create	Products loaded	Empty form, status label 「下書き」 (Draft), order number 「保存時に採番」 (Assigned on save)	Product options
Ready — edit (Draft)	Order + products loaded, status Draft	All inputs editable; status options 「下書き」/「進行中」/「取消」 (Draft/In progress/Cancelled)	Saved values

State	Trigger	UI behavior	Data shown
Ready — edit (locked)	Status InProgress/Completed/Cancelled	Product and quantity read-only with the lock hint 「下書き以外の製造指示では変更できません。」 (Can't be changed once the order leaves Draft); for Completed/Cancelled the status select is disabled too; due date and notes stay editable (DEC-004)	Saved values
Empty products	`GET /api/products` returns `[]`	Product select shows MSG-E014 and is disabled; Save shows MSG-E001	—
Saving	Save pressed, request in flight	Save disabled with label 「保存中…」 (Saving…); inputs stay as they are; Cancel still enabled	Entered values
Validation error	Client or 400 response	Inline errors (`FieldError` , `aria-invalid`), focus to the first invalid field	Entered values kept
Rule / conflict error	422 (MSG-E007/E008) or 409 (MSG-E009)	Error banner (`role="alert"`); for 409 the banner has a **再読み込み** (Reload) button (re-runs P-01 and discards local edits)	Entered values kept
Not found	404 on load	Panel MSG-E011 with a 「製造指示一覧に戻る」 (Back to production orders) link; no form	—
Forbidden	Missing role (client check) or 403	Panel MSG-E012 with a 「製造指示一覧に戻る」 (Back to production orders) link	—
Load error	Network/5xx on load	Panel MSG-E013 with **もう一度試す** (Try again) (re-runs P-01)	—
Success	201 / 200	Banner MSG-I001 / MSG-I002 (`role="status"`); form reset to the saved values, so `isDirty = false`	Saved values

Processing and state transitions

State transitions



Same-status updates and rejected transitions (422 MSG-E007) are in the table below; they don't change the state.

From state	Event	To state	Side effect
(none)	Create (P-02)	Draft	Row inserted; counter incremented; `orders.created` metric
Draft	Update with status InProgress	InProgress	Product/quantity become locked
Draft	Update with status Cancelled	Cancelled	Terminal
InProgress	Update with status Completed	Completed	Terminal; `completed_at_utc` set to the save's UTC time (WI-004 REQ-033)
InProgress	Update with status Cancelled	Cancelled	Terminal
any	Update with the same status	same	Other editable fields saved
any other pair	Update	unchanged	422 MSG-E007

Processing flows

The step-by-step processing is in the screen-processing companion, **001_DD-SPD** (`001_DD-SPD_製造指示登録・編集_20261024.md`), with one block per component:

Flow	Where	Events
P-01 Load	001_DD-SPD §1 `ProductionOrderPage`	E-01
P-02 Create / P-03 Update (client)	001_DD-SPD §2 `ProductionOrderForm`	E-02–E-06, E-08
P-02 / P-03 (server)	001_DD-FN §3 `CreateAsync`, §4 `UpdateAsync`; request boundary in 001_DD-SPD §5	—
P-04 Cancel	001_DD-SPD §4 `DiscardChangesDialog`	E-07, E-09
P-05 Save-response handling	001_DD-SPD §3	E-06

APIs used

Endpoint	Method	Purpose	Design doc
`/api/products`	GET	Product options (FN-004)	001_DD-API §1
`/api/production-orders`	POST	Create (FN-001)	001_DD-API §2
`/api/production-orders/{id}`	GET	Load (FN-003)	001_DD-API §3

Endpoint	Method	Purpose	Design doc
`/api/production-orders/{id}`	PUT	Update (FN-002)	001_DD-API §4

Database and transaction mapping

Operation	Table(s)	Transaction boundary	Concurrency handling
List products	`products` (`id, sku, name`)	none; `AsNoTracking`	not applicable
Load order	`production_orders` only (no join; the client resolves the product label from the product list)	none; `AsNoTracking` for GET	not applicable
Create	`production_order_number_counters` , `production_orders`	Explicit transaction: counter upsert + insert, then commit	Counter row lock; unique `(order_year, order_seq)` backstop
Update	`production_orders` (+ `products` existence query when `productId` changes); on `InProgress → Completed` the same `UPDATE` also writes `completed_at_utc` (003_DB)	Single `SaveChanges` (implicit transaction) — the status and the completion time commit together	`xmin` token (DEC-014): stale → 409

Queries select only the columns used (projection to DTOs). The runtime login is `pmai_app` , with the grants in 001_DB (DEC-016).

Exception handling

Failure	Retry policy	Timeout	User-facing error	Logging
Validation (400)	none	—	Inline field errors	Information: `ProductionOrderValidationFailed` with order id (if any), user id, failed field names (not values)
Business rule (422)	none	—	Banner MSG-E007/E008	Information: `ProductionOrderRuleViolated` with code, from/to status
Stale save (409)	none (user reloads)	—	Banner MSG-E009 + 「再読み込み」(Reload)	Information: `ProductionOrderConcurrencyConflict`
Not found (404)	none	—	Panel MSG-E011	Debug

Failure	Retry policy	Timeout	User-facing error	Logging
Counter overflow / DB error / unhandled	none automatic	Npgsql command timeout 30s (default)	Banner or panel MSG-E013	Error with exception (server-side only); response is generic Problem Details with no stack trace or exception text (`ai/rules/backend.md`)
Transient connection failure	EF Core `EnableRetryOnFailure` is **not** enabled: the create uses an explicit transaction, which would need an execution-strategy wrapper; not worth it for the demo	—	MSG-E013; user retries	Error
Frontend network failure	none automatic; user presses 「保存」 (Save) / 「もう一度試す」 (Try again)	browser default	MSG-E013	`console.error` only in development

Never logged: notes text, full request/response bodies, cookies.

Observability (OpenTelemetry)

Per `ai/rules/backend.md` and the design-consistency checklist. The backend has no OpenTelemetry setup yet, so plan revision 2 adds `OpenTelemetry.Extensions.Hosting` with ASP.NET Core and Npgsql instrumentation and the OTLP exporter. Npgsql spans cover the database statements; EF Core's own instrumentation package is still prerelease and the backend makes no outgoing HttpClient calls, so neither is added. The exporter is enabled only when `OTEL_EXPORTER_OTLP_ENDPOINT` is set; otherwise telemetry is collected but not exported.

Signal	Name	Attributes / tags	Emitted by
Span (auto)	`POST api/production-orders` , `PUT api/production-orders/{id}` , `GET ...`	`http.route` , `http.response.status_code`	ASP.NET Core instrumentation
Span (auto)	DB statements	`db.system=postgresql` (no parameter values)	Npgsql instrumentation
Span	`ProductionOrder.Create`	`production_order.id` , `production_order.number` , `outcome`	`ProductionOrderService` (ActivitySource `ProductionManagementAI.ProductionOrders`)
Span	`ProductionOrder.Update`	`production_order.id` , `status.from` , `status.to` , `outcome`	same
Counter	`pmai.production_orders.created`	`outcome` = `success` | `validation_failed` | `error`	Meter `ProductionManagementAI.ProductionOrders`
Counter	`pmai.production_orders.updated`	`outcome` = `success` | `validation_failed` | `rule_violation` | `conflict` | `not_found` | `error`	same

Signal	Name	Attributes / tags	Emitted by
Counter	`pmai.production_orders.status_transitions`	`from`, `to`	same, on successful status change
Histogram	`http.server.request.duration` (auto)	route, status	ASP.NET Core

The user id is not a span attribute (it's personal data). Logs correlate with spans through the trace id.

Configuration

Key	Default	Notes
`Plant:TimeZone`	`Asia/Tokyo`	IANA ID; validated at startup (DEC-017)
`ConnectionStrings:DefaultConnection`	—	Runtime login `pmai_app` (DEC-016); secret from environment
`OTEL_EXPORTER_OTLP_ENDPOINT`	unset	Optional

Accessibility (WCAG 2.2 AA)

- Every input has a `<label for>`. Required inputs have `aria-required="true"` and a visible `*` with the legend 「* は必須項目です。」 (* marks a required field).
- Errors: `aria-invalid="true"` + `aria-describedby` → the `FieldError` id. On a failed save, focus moves to the first invalid field.
- Read-only locked fields: native `readonly` on the quantity input. The product `<select>` can't be `readonly`, so it's rendered as a read-only text input showing the label, with `aria-describedby` → the lock hint. It stays focusable, unlike `disabled`.
- Banners: success `role="status"`, errors `role="alert"`. Panels get focus on render (`tabIndex=-1`).
- Language: `<html lang="ja">`, so assistive technology reads the Japanese labels with a Japanese voice (WI-005).
- Dialog: native `<dialog>` + `showModal()` (focus trap, Escape). It is labelled by its title via `aria-labelledby`.
- Focus-visible rings on every control; target size ≥ 24×24 CSS px (2.5.8); contrast ≥ 4.5:1 for text and ≥ 3:1 for control borders.
- Keyboard: the natural tab order matches the visual order; native date input and select.

Test viewpoints and unresolved decisions

Level: U = unit (xUnit Domain/Application, or Vitest+RTL for the frontend), I = integration (`WebApplicationFactory` + Testcontainers Postgres), E = E2E (Playwright, DEC-025). Test-plan IDs refer to `work-items/WI-002/test-plan.md` (TP-002).

Scenario	Precondition	Expected result	Test-plan ID
Create valid order (REQ-010) — I, E	Admin signed in, products seeded	201, `status=Draft`, `orderNumber` matches `^PO-\d{4}-\d{5}$`, redirect to edit + MSG-I001	TC-001
First order of a year gets `00001`; next gets `00002` (DEC-012/013) — I	Empty counter; plant clock faked to 2026 then 2027	`PO-2026-00001`, `PO-2026-00002`, `PO-2027-00001`	TC-002
Concurrent creates get distinct numbers — I	20 parallel POSTs	20 distinct sequential numbers, no gaps	TC-002
Year boundary uses the plant timezone (DEC-017) — U	`TimeProvider` at 2026-12-31T15:30Z (= 2027-01-01 00:30 JST)	Year 2027; "today" 2027-01-01	TC-003
Edit Draft order fields (REQ-011) — I	Draft order	200, values saved, `updatedAt` changed	TC-004
Unknown id (REQ-011) — I, U(fe)	—	404 MSG-E011; not-found panel	TC-005
Stale save (DEC-010) — I	Two clients load version v; A saves	B's PUT → 409 MSG-E009; row keeps A's values	TC-006
Unauthenticated (REQ-012) — I, U(fe)	No cookie	401 on all 4 endpoints; UI redirects to `/login`	TC-007
No role (REQ-012) — I, U(fe)	User without Admin/Operator	403 on all 4 endpoints; forbidden panel without API call	TC-008
Operator allowed (DEC-001) — I	Operator signed in	200/201	TC-009
Quantity rules (REQ-013) — U, I	—	`1`, `250`, `999999999` pass; empty, `0`, `-5`, `2.5`, `abc`, `1000000000` → MSG-E003/E010	TC-010
Due date rules (REQ-014, DEC-009) — U, I	Plant today = D	D and D+1 pass; D-1 → MSG-E005 on create; overdue order saved without changing the due date → 200; changing it to D-1 → MSG-E005	TC-011
Product rules (REQ-015) — I	—	Missing → MSG-E001; random GUID → MSG-E002	TC-012

Scenario	Precondition	Expected result	Test-plan ID
Notes rules (REQ-016, REQ-090) — U, I	—	Empty → `null`; 1000 chars (incl. emoji counted as 1) pass; 1001 → MSG-E006	TC-013, TC-470
Each allowed transition (REQ-017) — U, I	Order in the source state	200, new status; `allowedNextStatuses` updated	TC-014
Completion time recorded (WI-004 REQ-033) — U, I	InProgress order; clock pinned	Saving `Completed` sets `completed_at_utc` = the save time (= `updated_at_utc`); every other transition and every same-status save leaves it unchanged (`NULL`, or the original value on a later edit of a completed order's notes); a rejected save (422, 409) sets nothing; a request body carrying a completion field is ignored	WI-004 TP-004 (ID assigned in 003_DD)
Disallowed transitions (REQ-017) — U, I	e.g. Draft→Completed, Completed→InProgress, Cancelled→Draft	422 MSG-E007, status unchanged	TC-015
Status options per state (M-02) — U(fe)	Each status	Select options match; disabled for terminal states	TC-016
Locked-field change rejected whole (REQ-018, DEC-007) — U, I	InProgress order; PUT changes quantity + notes	422 MSG-E008; notes not saved	TC-017
Locked fields read-only in UI (REQ-018) — U(fe)	InProgress order	Product/quantity read-only, focusable, lock hint	TC-018
Cancel without changes (REQ-019) — U(fe), E	Pristine form	Navigates home, no dialog	TC-019
Cancel with changes → Discard / Keep editing / Escape (REQ-019) — U(fe), E	Edited form	Dialog; Discard → home; Keep editing/Escape → values kept, focus on Cancel	TC-020
Non-JSON body (DEC-020) — I	POST `text/plain` / form-encoded	415, nothing created	TC-021
Problem Details shape — I	Any 4xx/5xx	`application/problem+json`, `type`, `title`, `status`, `code`; no stack trace	TC-022
Runtime login privileges (DEC-016) — I	App connected as `pmai_app`	All flows work; `DELETE FROM production_orders` → permission denied	TC-023
Telemetry emitted — I	In-memory exporter	Create/Update spans + counters with the expected tags	TC-024
Accessibility — U(fe) + manual	—	Labels, `aria-invalid` / `describedby`, focus on first error, dialog focus; automated axe check (vitest-axe + @axe-core/playwright, DEC-026) has no violations	TC-025

Unresolved decisions: none. DD-level technical decisions DEC-021–DEC-024 are recorded in `work-items/WI-002/decisions.md`.